

Name: Seth.N, Richard.B, Niko.M**Date:** 3/4/2026**Class:** 1

MyDesign Portfolio

Element B

Element B: Documentation and Analysis of Prior Design Attempts

<1> Wrapping the Cable Around the Boom Pole

Image



(Figure 1) Cable Wrapped

Details

A common method used by students is manually wrapping the XLR cable around the boom pole to prevent it from hanging freely. However, the Audio Engineering Society (AES) states that tightly wrapping or bending audio cables repeatedly can cause internal conductor damage and reduce signal quality (Audio Engineering Society, 2018). Professional guidelines recommend avoiding sharp bends and stress points because they weaken cable shielding and internal wiring.

Audio Engineering Society. *Guidelines for Audio Cable Handling and Maintenance*. Audio Engineering Society, 2018, www.aes.org.

Analysis

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Strength:

- Keeps the cable out of the camera frame temporarily.
- Requires no additional materials or cost.

Shortcomings:

- Violates professional cable-handling guidelines.
- Increases long-term repair costs due to internal wire damage.
- Not adjustable once wrapped.
- Does not protect cable lifespan, which contradicts stakeholder needs for durability

This solution fails to meet the design requirements of being durable and protecting cables from damage. While it is inexpensive, it creates long-term costs and does not provide a sustainable or professional solution.

<2> Letting the Cable Hang Freely

Image



(Figure 2) Cable Hanging

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Another method is allowing the cable to hang freely from the microphone while recording. According to the National Association of Broadcasters (NAB), “proper cable management is essential to prevent safety hazards and maintain production quality.” (NAB, 2019). Loose cables increase the risk of tripping and can interfere visually with camera framing.

National Association of Broadcasters. *Best Practices for Audio Production and Cable Management*. NAB, 2019, www.nab.org.

Analysis

Strength:

- Added cost
- No stress applied to cable structure

Shortcomings:

- Frequently appears in camera shot
- Creates tripping hazards
- Does not maintain professional filming standards

This solution fails to meet the function of **holding cables up so they are not in shots**. It also does not align with the stakeholder's needs for safety and professional quality production

<3> Commercial Boom Pole Clips

Image



(Figure 3) Preexisting Clip

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Details

Professional boom pole clips are sold by audio equipment vendors like Sweetwater. These clips are designed to attach cables securely to microphone stands or poles. According to Sweetwater (2024), mic stand accessories improve cable organization and reduce interference during performances and recordings.

Sweetwater. *Mic Stands and Accessories Buying Guide*. Sweetwater Sound, 2024, www.sweetwater.com/insync/mic-stands-and-accessories-buying-guide/.

Analysis

Strength:

- Specifically designed for cable management.
- Keeps cables secure and straight.
- Professional and durable materials.

Shortcomings:

- Often sold in multi-packs at higher total cost.
- May exceed the target cost of \$2 per unit.
- Some models add noticeable weight to lightweight boom poles.

While this solution meets durability and effectiveness requirements, it may not meet the cost constraint for classroom budgets identified by stakeholders.

<4> Tape securing the Cable to the Boom

National Association of Broadcasters. *Best Practices for Audio Production and Cable Management*. National Association of Broadcasters, 2019, www.nab.org.

Image

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(Figure 4) Gaffer Tape

Some productions temporarily use gaffer tape to secure cables to equipment. According to NAB best practices, temporary fastening methods are sometimes used in live production settings (NAB, 2019). Tape can hold cables tightly against surfaces.

National Association of Broadcasters. *Best Practices for Audio Production and Cable Management*. National Association of Broadcasters, 2019, www.nab.org.

Analysis

Strength:

- Inexpensive per use.
- Lightweight.
- Keeps cable out of shot.

Shortcomings:

- Leaves adhesive residue over time.
- Not reusable.
- Adds recurring cost over time.
- Not adjustable once applied.
- Can damage cable insulation if removed improperly.

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This solution partially meets cost and weight objectives but fails in durability and long-term sustainability. It does not align with stakeholder needs for a reusable and adjustable design.

<5> Professional Cable Management Training and Maintenance

Details

The Bureau of Labor Statistics explains that audio technicians are responsible for maintaining and protecting equipment to extend its lifespan (BLS, 2022). Professional training emphasizes proper handling techniques and preventative maintenance rather than relying on damaging quick fixes.

Bureau of Labor Statistics. *Audio and Video Equipment Repairers: Occupational Outlook Handbook*. U.S. Department of Labor, 2022, www.bls.gov.

Analysis

Strength:

- Encourages proper care of equipment.
- Reduces long-term repair costs.
- Aligns with durability goals.

Shortcomings:

- Does not physically solve the problem of cables entering the shot.
- Requires training time.
- Does not provide a tangible device solution.

While training improves awareness, it does not meet the required function of physically holding cables to the boom pole. Therefore, it does not directly solve the defined problem.

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Work Cited

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Bureau of Labor Statistics. *Audio and Video Equipment Repairers: Occupational Outlook Handbook*. U.S. Department of Labor, 2022, www.bls.gov.

Figure 1, Bock, R. Cable Wrapped. 11 Feb, 2026. Author's Personal Collection.

Figure 2, Bock, R. Cable Hanging. 11 Feb, 2026. Author's Personal Collection.

Figure 3, Bock, R. Preexisting Clip. 11 Feb, 2026. Author's Personal Collection.

Figure 4, Wikipedia contributors. "Gaffer tape." *Wikipedia, The Free Encyclopedia*. Wikipedia, TheFree Encyclopedia, 18 Aug. 2025. Web. 4 Mar. 2026.

National Association of Broadcasters. *Best Practices for Audio Production and Cable Management*. National Association of Broadcasters, 2019, www.nab.org.

Sweetwater. *Mic Stands and Accessories Buying Guide*. Sweetwater Sound, 2024, www.sweetwater.com/insync/mic-stands-and-accessories-buying-guide/.

